SEC:: SR. JEE MAINS APEX | SPCL_GTM-04 (AN) | Date: 14.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
D	B	C	C	B	B	B	C	A	D
11	12	13	14	15	16	17	18	19	20
D	D	C	D	B	C	C	D	D	A
21	22	23	24	25					
2	16	13	1	16					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	C	B	D	C	C	B	B	B	A
36	37	38	39	40	41	42	43	44	45
A	C	C	D	C	B	C	D	D	B
46	47	48	49	50					
20	2	3	8	5					

MATHEMATICS

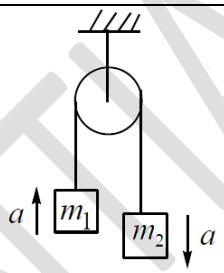
51	52	53	54	55	56	57	58	59	60
C	D	D	A	D	A	A	B	B	C
61	62	63	64	65	66	67	68	69	70
B	D	C	D	D	A	A	A	C	A
71	72	73	74	75					
55	13	75	221	806					

PHYSICS

1.	$\frac{PV^2}{PV} = C_V$ $\therefore C_P = C_V + R$
2.	$dB = \frac{\mu_0 i(dl)\sin\theta}{4\pi r^2}$ $= \frac{10^{-7} \times 10 \times 10^{-2} \times \sin 90°}{(0.5)^2}$ $= 4 \times 10^{-8}$ T
3.	$V_{rms} = \sqrt{\frac{3RT}{M}}$ $\therefore \frac{V_{rms}(He)}{V_{rms}(O)} = \sqrt{\frac{32}{4}}$ $= 2\sqrt{2}$.
4.	$E = \frac{1}{4\pi \in_0} . \frac{Q}{R^2}$ $Q = \sigma(4\pi R^2)$ $\therefore E = \frac{1}{4\pi \in_0} . \sigma\frac{(4\pi R^2)}{R^2}$ $E = \frac{\sigma}{\in_0}$.
5.	The photoelectric effect does not explain the wave nature of light.
6.	$F = kx$ $MLT^{-2} = KL$ DF of $[k] = MT^{-2}$ $\frac{\Delta K}{K} \times 100 = \frac{\Delta m}{m} \times 100 + 2\frac{\Delta T}{T} \times 100$ $= 1 + 2(2) = 5\%$
7.	$x_L = x_C$ $CSR = 42$ $LC = \frac{\text{Pitch of screw}}{\text{No. of H.S.D}}$ $LC = \frac{1}{100} = 10^{-2} mm$ $T.R = MSR + (LC)\ CSR = 1mm\ + 10^{-2}$ x 42 $T.R = 1.42mm$ $\frac{x}{50} = 1.42$ $x = 1.42$ x 50 $x = 71$
8.	$KE = \frac{1}{2} m\ Ve^2$ $= \frac{1}{2} m(2gR_E)$ $= mg\,R_E$

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9.	$V = u + at \Rightarrow 80 = 0 + at$ $\Rightarrow a = \frac{80}{t}$ $S_1 = \frac{1}{2}\left(\frac{80}{t}\right)t^2 = 40t$ $S_2 = vt = 80(3t)$ Average speed $= \frac{S_1 + S_2}{t_1 + t_2} =$ $= \frac{40t + 80(3t)}{4t}$ $= \frac{40t(1+6)}{4t} = 70$
10.	At resonance condition $x_L = x_C$. So minimum Impudence and maximum current in LCR series circuit.
11.	$a = \left(\frac{m_2 - m_1}{m_2 + m_1}\right)g$ $\Rightarrow \frac{g}{\sqrt{2}} = \left(\frac{m_2 - m_1}{m_2 + m_1}\right)g$ $\Rightarrow m_2 + m_1 = \sqrt{2}(m_2 - m_1)$ $\Rightarrow$ Divide both sides with m_2 $1 + \frac{m_1}{m_2} = \sqrt{2} - \sqrt{2}\frac{m_1}{m_2}$ $\Rightarrow \left(1 + \sqrt{2}\right)\frac{m_1}{m_2} = \sqrt{2} - 1 \Rightarrow \frac{m_1}{m_2} = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$
12.	E=2.48 eV $\phi_0 = V_0 = 0.5\,V$ $KE_{max} = eV_0 = 0.5\,eV$ According to Einstein PE equation KE $_{max}$ = E $_{incident}$ - ϕ_0 $\phi_0 = 2.48 - 0.5 = 1.98$ eV
13.	$m = 50g = 50 \times 10^{-3}\,Kg$ $u = 100\,m/s$, $V = 40\,m/s$ Percentage loss in $KE = \frac{KE_i - KE_f}{KE_i} \times 100$ $= \frac{\frac{1}{2}mu^2 - \frac{1}{2}mv^2}{\frac{1}{2}mu^2} \times 100$ $= \left(1 - \frac{v^2}{u^2}\right) \times 100 = \left[1 - \left(\frac{40}{100}\right)^2\right] \times 100$ $= (1 - 0.16)100 = 84\%$

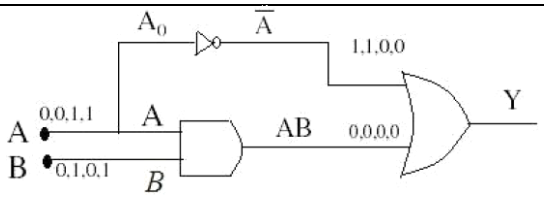


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14. $KE = \frac{p^2}{2m} \Rightarrow KE\ \alpha \frac{1}{m}$

Lowest mass of A,B,C,D is $\frac{m}{2}$

$\Rightarrow E_{max}$ for A

15. 

$Y = AB + \bar{A}$

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

16. Volume of the ball, $V = \frac{m}{\rho}$, mass of liquid displaced by the ball. 'ρ' is density of ball.

When the ball falls with a constant velocity,

F_V = Effective weight of the ball

$F_V = mg - m'g = g(m - m')$

$= \left(m - \frac{m\rho_0}{\rho}\right)g$

$\therefore F_V = mg\left(1 - \frac{\rho_0}{\rho}\right)$

17. (a) Torque $(\tau) = F \times \perp^r$ distance $= \left[ML^2T^{-2}\right]$

(b) Magnetic field (B) $= \frac{F}{m} = \frac{MLT^{-2}}{IL} = \left[MI^{-1}T^{-2}\right](or)\left[MT^{-2}A^{-1}\right]$

(c) Magnetic moment (M) = $2lm$ = $L \times IL = \left[IL^2\right](or)\left[L^2A^1\right]$

(d) $\mu_0 = \frac{Fr^2}{m_1m_2} = \frac{MLT^{-2} \times L^2}{I^2L^2} = \left[MLT^{-2}A^{-2}\right]$ $\quad (\because I = A)$

18. Speed of light $c = 3 \times 10^8 m/s$

Speed of light in medium $v = 1.5 \times 10^{-8} m/s$

$v = \frac{1}{\sqrt{\mu_0\varepsilon_0}.\sqrt{\mu_r\varepsilon_r}} = 1.5 \times 10^8 m/s;\ \mu_r = 2$

$\varepsilon_r = 2$

19. For Balmer series, $n_1 = 2, n_2 = \infty$ $\quad \therefore \lambda_{LS} = \frac{1}{R}$ .....(2)

$\frac{1}{\lambda_{BS}} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = R\left[\frac{1}{2^2} - \frac{1}{\infty^2}\right] = \frac{R}{4}$

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$\therefore \lambda_{BS} = \frac{4}{R}$ .....(1)

For Lyman series, $n_1 = 1, n_2 = \infty$

$$\frac{1}{\lambda_{LS}} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = R\left[\frac{1}{1} - \frac{1}{\infty}\right] = \frac{R}{1}$$

$$\frac{eqn(1)}{eqn(2)}\ \frac{\lambda_{BS}}{\lambda_{LS}} = \frac{4}{R} \times \frac{R}{1} = 4:1$$

20.

Between $A \& B \rightarrow P = \frac{24 \times 24}{48} = 12\,\Omega$

$B \& C \rightarrow Q = \frac{1 \times 1}{(1+1)} = \frac{1}{2} = 0.5\ \Omega$

$A \& D \rightarrow R = (6 + x)\ \Omega$

$C \& D \rightarrow S = 0.5\ \Omega$

A/c to wheatstone's Bridge principle,

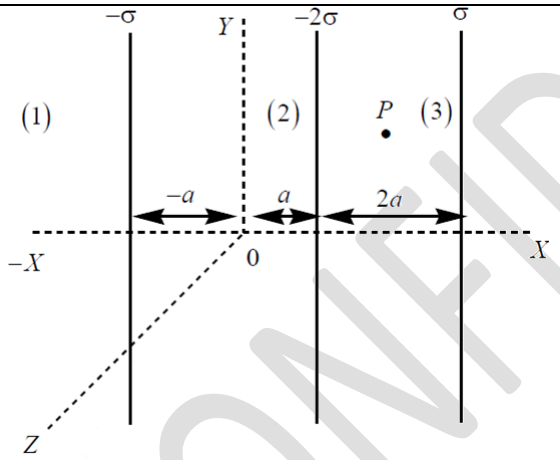
$$\frac{P}{Q} = \frac{R}{S}$$

$$\frac{12}{0.5} = \frac{6+x}{0.5}$$

$12 = 6 + x$

$\therefore\ x = 6\,\Omega$

21.



Electric field due to 1st sheet is $\frac{\sigma}{2\varepsilon_0}$ on left side −(1)

Electric field due to 2nd sheet is $\frac{2\sigma}{2\varepsilon_0}$ on left side −(2)

Electric field due to 3rd sheet is $\frac{\sigma}{2\varepsilon_0}$ on left side −(3)

$\therefore$ Total electric field is

$$E = \frac{\sigma}{2\varepsilon_0} + \frac{2\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{4\sigma}{2\varepsilon_0} = \frac{2\sigma}{\varepsilon_0} = 2$$

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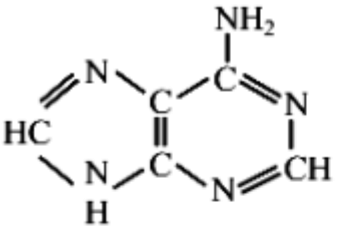
	Electric field due to 1st sheet is $\frac{\sigma}{2\varepsilon_0}$ on left side $-(1)$ Electric field due to 2nd sheet is $\frac{2\sigma}{2\varepsilon_0}$ on left side $-(2)$ Electric field due to 3rd sheet is $\frac{\sigma}{2\varepsilon_0}$ on left side $-(3)$ $\therefore$ Total electric field is $E=\frac{\sigma}{2\varepsilon_0}+\frac{2\sigma}{2\varepsilon_0}+\frac{\sigma}{2\varepsilon_0}=\frac{4\sigma}{2\varepsilon_0}=\frac{2\sigma}{\varepsilon_0}=2$ Electric field due to 1st sheet is $\frac{\sigma}{2\varepsilon_0}$ on left side $-(1)$ Electric field due to 2nd sheet is $\frac{2\sigma}{2\varepsilon_0}$ on left side $-(2)$ Electric field due to 3rd sheet is $\frac{\sigma}{2\varepsilon_0}$ on left side $-(3)$ $\therefore$ Total electric field is $E=\frac{\sigma}{2\varepsilon_0}+\frac{2\sigma}{2\varepsilon_0}+\frac{\sigma}{2\varepsilon_0}=\frac{4\sigma}{2\varepsilon_0}=\frac{2\sigma}{\varepsilon_0}=2$
22.	Same wide is stretched $\therefore$ Volume is constant $\therefore$ From $R=\frac{\rho l}{A}$ $R=\frac{\rho l}{A}\times\frac{A}{A}$ $R=\frac{\rho V}{A^2}$ $R\ \alpha\frac{1}{A^2}$ $R\ \alpha\frac{1}{\left(\pi r^2\right)^2}$ $R\ \alpha\frac{1}{\pi^2 r^4}$ $R=\frac{1}{r^4}$ $\frac{R_1}{R_2}=\frac{r_2^4}{r_1^4}$ $R_2=16R$
23.	$\frac{2}{5}\pi R^2 w=\frac{2}{5}\pi\left(\frac{3R}{4}\right)^2 w_1$ $w_1=\frac{16w}{9}$ $T_1=\frac{2\pi}{w_1}=\frac{2\pi}{w}\times\frac{9}{16}=24\times\frac{9}{16}hours$ $T_1=13\ hours\ 30\ \text{min}.$

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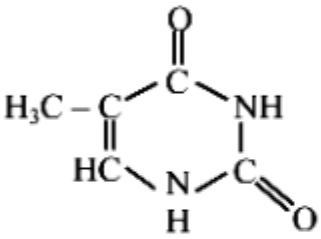
	$T_1 = 13\ hours\ 30\ \text{min}.$ $T_1 = 13\ hours\ 30\ \text{min}.$
24.	By conservation of volume $\frac{\cancel{4}}{\cancel{3}}\pi R^3 = 1000 \times \frac{\cancel{4}}{\cancel{3}}\pi r^3$ $R^3 = 1000r^3 \Rightarrow r = \frac{R}{10}$ Surface energy of 1000 droplets $= 1000 \times T \times 4\pi\left(\frac{R}{10}\right)^2 = 10\left(T \times 4\pi R^2\right)$ Surface energy of big drop $= T \times 4\pi R^2$ $\frac{Surface\ energy\ of\ 1000\ droplets}{Surface\ energy\ of\ big\ drop} = \frac{10\left(T \times 4\pi R^2\right)}{T \times 4\pi R^2}$ $\frac{10}{x} = \frac{10}{1}$ $\therefore x = 1$
25.	$\gamma_n = 0.53\frac{n^2}{z} \cdot A^0$ $8.48A^0 = 0.53 \times \frac{n^2}{1}A^0$ $n^2 = \frac{8.48}{0.53} = 16$ $n = 4$ $E = -13.6 \times \frac{Z}{n^2}$ $E_4 = \frac{-13.6}{16} = 0.85ev$ $\frac{E}{x} = \frac{-13.6}{16} = 0.85ev$ $x = 16$

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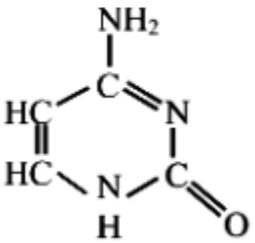
26.



⇒ Adenine



⇒ Thymine



⇒ Cytosine

Are bases of DNA molecule. As DNA contain four bases, which are adenine, guanine, cytosine and thymine.

27. Cm – Curium is not lanthanoid
Cm – is Actinoid.

28. Graphite is conductor due to presence of mobile π electrons.

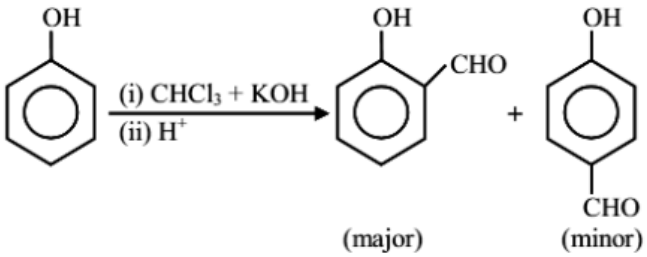
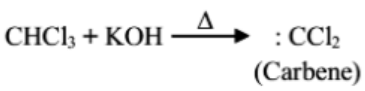
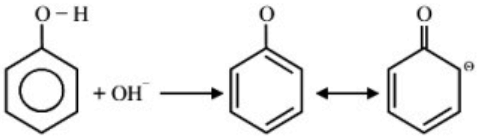
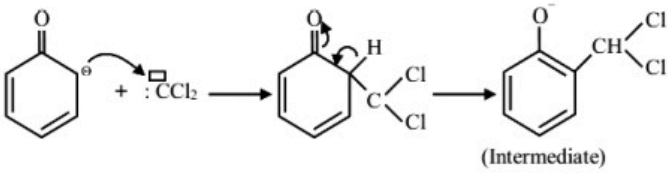
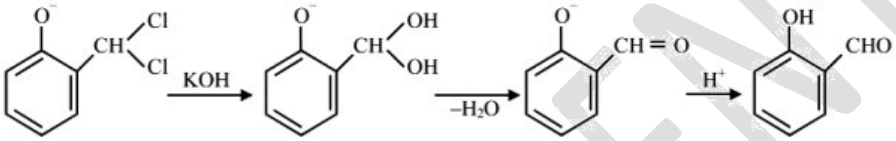
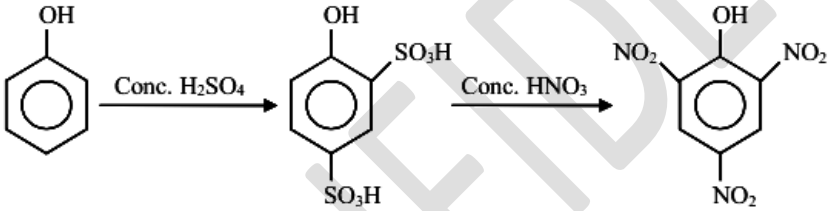
29. $$m = \frac{1000 \text{ x M}}{1000\,\text{xd} - \text{M x M.wt of solute}}$$

$m = 3$

M.wt of NaOH = 40

$d = 1.12 \text{ g mL}^{-1}$

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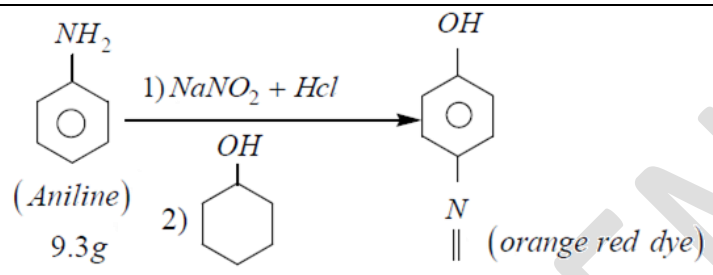
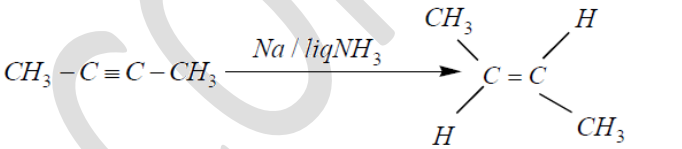
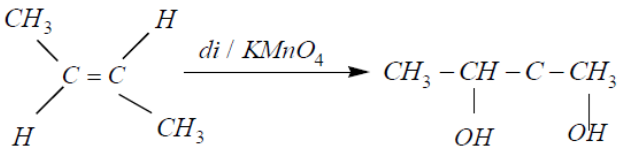
30.	Reimer Tiemann Reaction  (major) (minor) Mechanism -  (Carbene)   (Intermediate) 
31.	Picric acid – 2, 4, 6 – trinitrophenol  (Phenol –2, 4 – disulphonic acid) (Picric acid or 2, 4, 6 – Trinitrophenol)
32.	Group – I Cations - Pb^{+2}, Group reagent – dil HCl Group – III Cations Fe^{+3}, Cr^{+3}, Group reagent - $NH_4Cl + NH_4OH$ Group – IV Cations - $CO^{+2}, Ni^{+2}, Zn^{+2}, Mn^{+2}$, Group reagent - $NH_4OH + NH_3Cl + H_2S$ Group – V Cations - $Ca^{+2}, Sr^{+2}, Ba^{+2}$, Group reagent - $NH_4OH + (NH_4)_2CO_3$
33.	*Ga* – used in thermometers to measure temperature range 302K to 2673k.
34.	Metamers are the isomers having same functional group but difference in Alkyl (or) aryl groups attached to two sides of functional groups
35.	– SO_3H (Sulphonic acid group)
36.	

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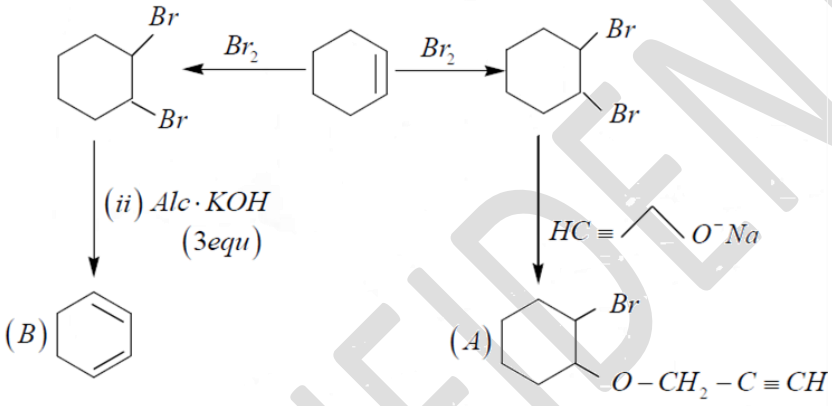
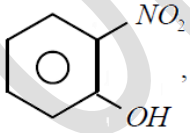
	$\alpha = \sqrt{\frac{K_a}{c}} = \sqrt{\frac{1.2\times10^{-5}}{0.03}} = 2\times10^{-2}$ $= 0.02$ $i = 1 + (n-1)\alpha = 1 + (2-1)0.02$ $= 1.02$ $\pi = iCRT = 1.02 \times 0.03 \times 0.083 \times 300$ $= 0.76194$ $= 76.194 \times 10^{-2}$
37.	Conceptual
38.	Formaldehyde is more reactive towards nucleophilic addition due to less electro density and less steric hindrance on Carbonyl carbon.
39.	No. of molecules same then No. of moles also same. for $\Delta ng = 0$ $K_p = K_C$ $K_p = K_c = \frac{(\frac{n}{v})^2}{(\frac{n}{v})(\frac{n}{v})} = 1$ $K_p = 10 \times 10^{-1}$
40.	(A) $SO_2.Cl_2$ No. of electron pairs $= \frac{V+M-C+A}{2} = \frac{6+2-0+0}{2} = 4$ Tetrahedral (B) NO = It has $7+8 = 15e^-$ paramagnetic (C) $NO_2^- = 7+8+8+1 = 24e^-$ diamagnetic (D) $I_3^- = \frac{7+2-0+1}{2} = 5$ 2L.P.'s. 3 B.P's sp^3d hybridization Linear Geometry
41.	Electron pairs $= \frac{V+M-C+A}{2}$ sp^3 = Tetrahedral dsp^2 = Sqaure planar sp^3d = Trigonal Bipyramidal sp^3d^2 = Octahedral
42.	Electron pairs $= \frac{V+M-C+A}{2}$ SF_4 – $\frac{6+4-0+0}{2} = 5$ sea saw BrF_3 sp^3d - 3 B.P's 2 L.P's Bent-T-Shape BrO_3^- sp^3 - 3 B.P's 1 L.P's Pyramidal shape NH_4^+ sp^3 - 4 B.P's Tetrahedral

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43.	Iodoform – Used as Antiseptic CCl_4 – Fire extinguisher CFCs – Refrigerant DDT – Insecticide
44.	Strength of ligand is $\alpha \frac{1}{\text{Wave length}}$ And directly proportional to wave number. CN^- is strong field and H_2O is weak field. $\therefore \left[Co(CN)_6\right]^{3-} > \left[Co(NH_3)_5(H_2O)\right]^{3+} > \left[CoCl(NH_3)_5\right]^{2+} > \left[Cu(H_2O)_4\right]^{2+}$
45.	The solution is already diluted, so further dilution no change in molar conductivity, i.e. Conductivity remains same.

46.	 $93g \rightarrow 198g$ $9.3g \rightarrow ?$ $= \frac{198 \times 9.3}{93} = 19.8 = 20$
47.	 *Trans – 2 – butene*  *butane – 2,3 – diol*
48.	99.9% completion,

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	$[A_0]=100 \quad \Rightarrow 90\%\ completion$ $[A]=100-99.9=0.1 \Rightarrow [A_0]=100$ $K=\frac{2.303}{t}\log\frac{[A_0]}{[A]} \quad \Rightarrow [A]=100-90=10$ $t=\frac{2.303}{k}\log\frac{100}{0.1}-(1) \Rightarrow t=\frac{2.303}{k}\log\frac{100}{90}-(2)$ $\frac{t_1}{t_2}=\frac{\frac{2.303}{t}\log\frac{100}{0.1}}{\frac{2.303}{k}\log\frac{100}{10}}$ $=\frac{\log 1000}{\log 10}$ $=3\log 10=3$ $t_1=3t_2$ or Shortcut method $t_{99.9\%}=3t_{90\%}=3$
49.	 $A=4,\ B=4$ $\therefore A+B=8$
50.	The molecules can exhibit hydrogen bonding are: 5 $CH_3OH, H_2O,$  $, HF, NH_3$ Compounds can form hydrogen bonding.

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51. $n = 20$, Given $\sum \frac{x_i}{n} = 10$

$$\sum x_i = 20x10 = 200$$

$$\boxed{\text{Corrected mean} = \frac{204}{20} = 10.2}$$

Given S.D = 2

$$2 = \sqrt{\frac{\sum x_i^2}{n} - \left(\bar{x}\right)^2}$$

$$4 = \frac{\sum x_i^2}{20} - (10)^2$$

$$20 \text{ x } 104 = \sum xi^2$$

$$\sum x_i^2 = 2080$$

Corrected $\sum x_i^2 = 2080 - 8^2 + 12^2 = 2160$

$$\text{Corrected S.D} = \sqrt{\frac{\sum x_i^2}{n} - \left(\bar{x}\right)^2}$$

$$= \sqrt{\frac{2160}{20} - (10.2)^2}$$

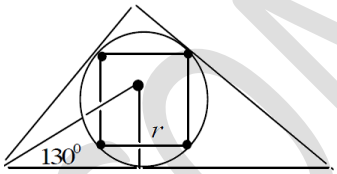
$$= \sqrt{108 - 104.04}$$

$$\text{S.D} = \sqrt{3.96}$$

52. $\tan 38 = \frac{r}{6}$

$$\frac{1}{\sqrt{3}} = \frac{r}{6} \Rightarrow \sqrt{2}r$$

$$\boxed{r = 2\sqrt{3}}$$



Side of square = $\sqrt{2}r$

$$= \sqrt{2}.2\sqrt{3}$$

$$= 2\sqrt{6}$$

Area = $(2\sqrt{6})^2 = 4 \text{ x } 6 = 24 = m$

Primate = $4 \text{ x } 2\sqrt{6} = 8\sqrt{6} = n$

$$m + n^2 = 24 + \left(8\sqrt{6}\right)^2$$

$$= 24 + 64 \text{ x } 6$$

$$= 24 + 384$$

$$= 408.$$

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53.

$$\int_0^{\pi/4} \frac{\sin^2 x \cos^2 x}{\left(\sin^3 x + \cos^3 x\right)^2} dx$$

$$= \int_0^{\pi/4} \frac{\sin^2 x \cos^2 x}{\cos^6 x \left(1 + \tan^3 x\right)^2} dx$$

$$= \int_0^{\pi/4} \frac{\tan^2 x \sec^2 x dx}{\left(1 + \tan^3 x\right)^2} dx$$

$$= \int_1^2 \frac{1}{3} \frac{dt}{t^2} \qquad \text{Pit } 1 + \tan^3 x = t$$

$$3 \tan^2 x . \sec^2 x dx = dt$$

$$= -13\left(\frac{1}{t}\right)_1^2 = -\frac{1}{3}\left(\frac{1}{2} - 1\right)$$

$$= -\frac{1}{3}\left(-\frac{1}{2}\right) = \frac{1}{6}$$

54.

P (Standard automobile from A) = $\frac{6}{10}$ x $\frac{8}{10}$ = $\frac{12}{25}$

P (Standard automobile from B) = $\frac{4}{10}$ x $\frac{9}{10}$ = $\frac{9}{25}$

P (Selected standard automobile combination from B) =

$$P = \frac{7/25}{\frac{12}{25} + \frac{9}{25}} = \frac{9}{21} = \frac{3}{7}$$

$$\therefore 126P = 126 \text{ x } \frac{3}{7} = 54$$

55.

Area of the quadrilateral formed by points $A(3,1,-1)$, $B\left(\frac{5}{3}, \frac{7}{3}, \frac{1}{3}\right)$, $C(2,2,1)$, and $D\left(\frac{10}{3}, \frac{2}{3}, \frac{-1}{3}\right)$ is $\frac{1}{2}\left|\overrightarrow{AC} \text{ X } \overrightarrow{BD}\right|$

Now $\overrightarrow{AC}$ X $\overrightarrow{BD}$ = $\begin{vmatrix} u & j & l \\ -1 & 1 & 2 \\ \frac{5}{3} & \frac{-5}{3} & \frac{-2}{3} \end{vmatrix}$

$$= \frac{8}{3}i + \frac{8}{3}j$$

Area = $\frac{1}{2}\left|\frac{8}{3}i + \frac{8}{3}j\right| = \frac{1}{2} \frac{8}{3}\sqrt{2} = \frac{4}{3}\sqrt{2}$

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56. Equation of the line passing through $(4,-9)$ and having style 'm' is

$y+9=m(x-4)$

$mx-y-(4m+9)=0$

$$\frac{x}{\frac{4m+9}{m}}+\frac{y}{-(4m+9)}=1$$

Sum of distance from origin be

$$f(m)=OA+OB=\frac{4m+9}{m}+4m+9$$

$$f(m)=\frac{4m^2+13,+9}{m}=4m+13+\frac{9}{m}$$

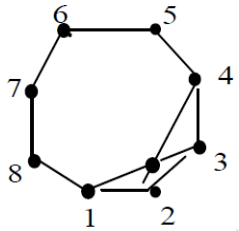
$$f'(m)=4-\frac{9}{m^2}=0$$

$$\Rightarrow m^2=\frac{9}{4}\Rightarrow m=\pm\frac{3}{2}$$

$$f''(m)=\frac{18}{m^3}>0 \text{ if } m=\frac{3}{2}$$

$$\therefore \text{ minimum } = f\left(\frac{3}{2}\right)=4\left(\frac{3}{2}\right)+13+\frac{9}{3/2}=25$$

57. Total triangles $={}^8C_3=\frac{8.76}{32.1}=56$



Triangles with two sides common are 1,2, 3, 2, 3, 4 ,…..8, 1, 2 = 8

Triangles with three sides common = 0.

Triangles with one side common $={}^8C_1.{}^4C_1=32$

Total triangles with no sides common with sides of octagon =

(total) – [with + side common + two sides common + three sides common]

= 56 – (32 + 8 + 0) = 16

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58.

$$A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-1 & 3 & \frac{n}{2}(3n-1) \end{vmatrix}$$

$C_1 : C_1 - rC_2$

$$A_r = \begin{vmatrix} 0 & 1 & \frac{n^2}{2}+\alpha \\ 0 & 2 & n^2-\beta \\ -2 & 3 & \frac{3n^2}{2}-\frac{n}{2} \end{vmatrix}$$

$$A_r = -2\left[n^2-\beta-2\left(\frac{n^2}{2}+\alpha\right)\right]$$

$$A_r = (\beta+2\alpha)2$$

$$A_r = 2\beta+4\alpha$$

59.

$$\vec{b_1} X \vec{b_2} = \begin{vmatrix} i & j & k \\ 2 & -7 & 5 \\ 2 & 1 & -3 \end{vmatrix} = 16(i+j+k)$$

$$\left|\vec{b_1} X \vec{b_2}\right| = 16\sqrt{3}$$

$$\left(\vec{a_2}-\vec{a_1}\right).\left(\vec{b_2}-\vec{b_1}\right) = (-4,16,0).16(1,1,1)$$

$$\left(\vec{a_2}-\vec{a_1}\right).\left(\vec{b_2}-\vec{b_1}\right) = (-4,16,0).16(1,1,1)$$

$$= 16(-4+16) = 16\text{x}12$$

$$\text{S.D} = \frac{16\text{x}12}{16\sqrt{3}} = 4\sqrt{3}$$

60.

$$\cot^{-1}3+\cot^{-1}4+\cot^{-1}5+\cot^{-1}n = \frac{\pi}{4}$$

$$\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{4}+\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{n} = \frac{\pi}{4}$$

$$\tan^{-1}\frac{7}{11}+\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{n} = \frac{\pi}{4}$$

$$\tan^{-1}\frac{46}{48}+\tan^{-1}\frac{1}{n} = \tan^{-1}1$$

$$\tan^{-1}\frac{1}{n} = \tan^{-1}\frac{1}{1}-\tan^{-1}\frac{23}{24}$$

$$= \tan^{-1}\frac{1}{47}$$

$$\therefore n = 47$$

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61. $f(x)=x^x$

$f'(x)=x^x(1+\log x)>0$

$=1+\log x>0$

$\log x>-1$

$\Rightarrow x>\frac{1}{e}\Rightarrow\left(\frac{1}{e},\infty\right)$

62. $\frac{dy}{dx}+\frac{y}{1+x^2}=\frac{e^{Tan^{-1}x}}{1+x^2}$

I.F $=e^{\int\frac{1}{1+x^2}dx}=e^{Tan^{-1}x}$

$y.e^{Tan^{-1}x}=\int\frac{\left(e^{Tan^{-1}x}\right)^2}{1+x^2}dx$ Put $\tan^{-1}x=t$

$\frac{1}{1+x^2}dx=dt$

$y.e^{T\,an^{-1}x}=\frac{e^{Tan^{-1}x}}{2}+d$

$x=0,y=0\Rightarrow 0=\frac{e^{\frac{\pi}{2}}}{2}+c\Rightarrow e=-\frac{1}{2}e^{\pi/2}$

$ye^{Tan^{-1}x}=\frac{e^{2Tan^{-1}x}}{2}-\frac{e^{\pi/2}}{2}$

$x=0\Rightarrow y=\frac{1}{2}-\frac{e^{\pi/2}}{2}=\frac{1}{2}\left(1-e^{\pi/2}\right)$

63. $f(x)=\frac{(x+5)(x-3)}{x^2-4x+9}$

$\Rightarrow f(3)=f(5)\Rightarrow f$ is not one-one

for $x^2-4x+9,\Delta=b^2-4ac<0\Rightarrow f$ is not onto

64. Given equation $x^2-\left(t^2-5t+6\right)x+1=0$

$\therefore\ \alpha_{2025}-\left(t^2-5t+6\right)\alpha_{2024}+\alpha_{2023}=0$

$\frac{\alpha_{2025}+\alpha_{2023}}{\alpha_{2024}}=t^2-5t+6$

$=t^2-5t+\frac{25}{4}+6-\frac{5}{4}$

$=\left(t+\frac{5}{2}\right)^2+\left(-\frac{1}{4}\right)\geq 0$

Min.value $=\frac{-1}{4}$

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65. R is defined on $X=\{1,2,3,....20\}$

$R_1=\{(x,y):2x-3y=2$

$R_2=\{(x,y):5x+4y=0$

So, $R_1=\{(4,3),(7,4),(10,6),(13,8),(16,10),(19,12)\}$

for symmetric we need to add 6 elements

here as $\{(3,4),(4,7),(6,10),(8,13),(10,16),(12,19)\}$

So y = 6

For $R_2=5x-4y=0$

$R_2=\{(4,5),(8,10),(12,15),(10,20)\}$

for symmetric we need to add 4 elements

$\{(5,4),(10,8),(15,12),(20,16)\}$

$N=4 \qquad \Rightarrow M+N=10$

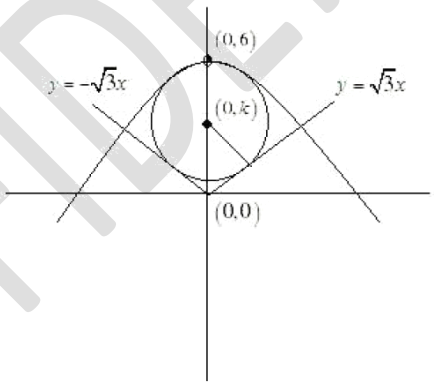
66. $y=6-x^2$

$x^2=6-y$

$x^2=-(y-6)$ ...(1)

$(x-0)^2=-4a(y-6)$

(0,6)



Centre of circle (0,k)

Radius r = 6 – k

$6-k=\left|\frac{0-k}{\sqrt{3+1}}\right|=\frac{k}{2}$

$6-k=\frac{k}{2}$

$12-2k=k$

$3k=12$

$k=4$

$y=\sqrt{3}x$

Centre (0,4) $r=6-4=2$

Equation of circle $(x-0)^2+(y-4)^2=4$

$x^2+y^2-8y+16=4$

$x^2+y^2-8y+12=0$

Option : 1 (2,4)

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67. $\frac{dy}{dx}+\frac{y}{x\log_e x}=\frac{3}{2x^2}$

$I.F=e^{\int\frac{1}{x\log_e x}}=e^{\log(\log_e x)}=\log_e x$

$y\log_e x=\frac{3}{2}\int\frac{\log_e x}{x^2}\,dx$

$y\log_e x=\frac{3}{2}\int t\,e^{-t}\,dt \qquad \log_e x=t$

$=\frac{-3}{2}\frac{1}{x}(\log_e x+1)+C$

$x=e^{-1}, y=0\Rightarrow 0=-\frac{3}{2x}(-1+1)+c$

$y\log_e x=\frac{-3}{2x}(\log_e x+1)$

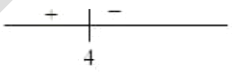
$x=e\Rightarrow y=\frac{-3}{2e}(1+1)=\frac{-3}{e}$

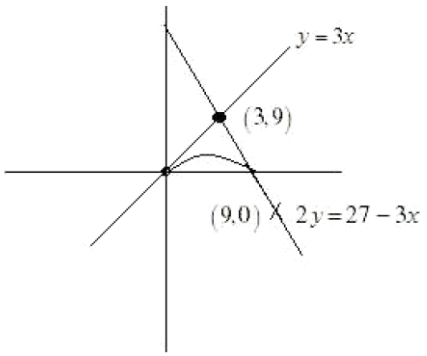
68. A = {100,101,102,.....700}
n(multiple of 3) = 233 – 33 = 200
n(multiple of 4) = 175 – 24 = 151
n(multiple of 3 & 4) = 58 – 8 = 50
n(neither multiple of 3 nor multiple of 4) = 601 – 301 = 300

69. $y=3x, 2y=27-3x, y=3x-x^{3/2}$

$y=3x-x^{3/2}$

$y'=3-\frac{3}{2}\sqrt{x}\Rightarrow f'(x)$





$\text{Area}=\int_0^3 3x-\left(3x-x^{3/2}\right)dx$

$+\int_3^9\frac{27-3x}{2}-\left(3x-x^{3/2}\right)dx$

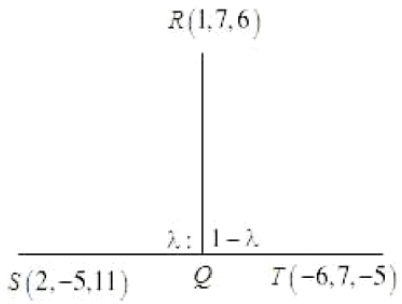
$=\int_0^3 x^{3/2}dx+\frac{1}{2}\int_3^9(27-3x)dx-\int_3^9 3x\,dx+\int_3^9 x^{3/2}dx$

$=\frac{2}{5}\left(x^{5/2}\right)_0^3+\frac{1}{2}\left(27x-\frac{3}{2}x^2\right)_3^9-3\left(\frac{x^2}{2}\right)_3^9+\frac{C}{5}\left(x^{3/2}\right)_3^9$

$=\frac{81}{5}\Rightarrow 10A=10\times\frac{81}{5}=162$

70. $f(x)=x^3\sin\left(\frac{1}{x}\right)$

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	$f'(x)=x^3\cos\left(\frac{1}{x}\right)\left(\frac{-1}{x^2}\right)+3x^2\sin\left(\frac{1}{x}\right)$ $f'(x)=-x\cos\left(\frac{1}{x}\right)+3x^2\sin\left(\frac{1}{x}\right)$ $f''(x)=\frac{-x\sin\left(\frac{1}{x}\right)}{x^2}-\cos\left(\frac{1}{x}\right)+3x^2\cos\left(\frac{1}{x}\right)\left(\frac{-1}{x^2}\right)+6x\sin\left(\frac{1}{x}\right)$ $=\frac{-\sin\left(\frac{1}{x}\right)}{x}-4\cos\left(\frac{1}{x}\right)+6x\sin\left(\frac{1}{x}\right)$ $f''(0)$ is not defined $f''\left(\frac{2}{\pi}\right)=-\frac{\pi}{2}+6.\frac{2}{\pi}=\frac{24-\pi^2}{2\pi}$
71.	Given $\alpha x+y+2z=0$ $x+y\beta+3z=0$ $2x+2y+z\gamma=0$ Have nontrivial solution (x,y,z) $\neq(0,0,0)$ $\Rightarrow\begin{vmatrix}\alpha & 1 & 2\\ 1 & \beta & 3\\ 2 & 2 & \gamma\end{vmatrix}=0\Rightarrow\alpha(\beta\gamma-b)-1(\gamma-b)+2(2-2\beta)=0$ $\Rightarrow\alpha\beta\gamma-b\alpha-\gamma+6+4-4\beta=0$ $\Rightarrow 45-6\alpha-4\beta-\gamma+10=0$ $\therefore\ 6\alpha+4\beta+\gamma=55$
72.	Given P(10,–2,–1), S(2,–5,11), T(–6,7,–5) Let $SQ:QT=\lambda:1-\lambda$ $\therefore Q(2-8\lambda,12\lambda-5,5-16\lambda)$ DRs of QR : $1-8\lambda,12\lambda-12,5-16\lambda$ DRs of ST : 2,-3,4 Given $QR\perp ST$ $\Rightarrow Q(1-8\lambda)-3(12\lambda-12)+4(5-16\lambda)=0$ $\Rightarrow\lambda=\frac{1}{2}$ $\therefore Q(-2,1,3)$ $\therefore PQ=\sqrt{144+9+16}=\sqrt{169}=13$ 
73.	Q = (4, – 2) P = (x, y)

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	Slop of test=$\frac{1}{2}\left(\frac{y+5}{x-3}\right)$ $\frac{dy}{dx}=\frac{1}{2}\left(\frac{y+5}{x-3}\right)$ $\frac{2dy}{y+5}=\frac{dx}{x-3}$ $(y+5)^2=c(x-3)$ pass through $(4,-2)$ $9=C(1)\Rightarrow C=9$ $(y+5)^2=9(x-3)$ $A=(3,-5),\ 4a=9, a=\frac{9}{4}$ $S=\left(3+\frac{9}{4},-5\right)=\left(\frac{21}{4},-5\right)$ $R=(7,1)$ $d=SR=\sqrt{\frac{49}{16}+36}$ $=\sqrt{\frac{625}{16}}=\frac{25}{4}$ $12d=12\times\frac{25}{4}=75$
74.	$4x^4-17x^2+8x^3-12x+9=0$ Replace x by $\sqrt{x}$ $4x^2+8x\sqrt{x}-17x-12\sqrt{x}+9=0$ $\sqrt{x}(8x-12)=-(4x^2-17x+9)$ $\sqrt{x}(8x-12)=-4x^2+17x-9$ SBS $x(8x-12)^2=(-4x^2+17x-9)^2\ x(64x^2+144-192x)$ $=16x^4+289x^2+81-136x^3-360x+72x^2$ $64x^3+144x-192x^2=16x^4+289x^2+81-136x^3-306x+72x^2$ $16x^4-200x^3+553x^2-450x+81=0$ Replace x by x-4

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	$16(x-4)^4-200(x-4)^3+553(x-4)^2-450(x-4)+81=0$ $S_4=\frac{256\times16+12800+553\times10+450\times4+81}{16}$ $=\frac{4096+12800+8848+1800+81}{16}$ $=\frac{27625}{16}=\frac{125}{16}\times221$ $m=221$
75.	$T_2=n_{c_1}x^{n-1}y=135\rightarrow(1)$ $T_3=n_{c_2}x^{n-2}y^2=30\rightarrow(2)$ $T_4=n_{c_3}x^{n-3}y^3=\frac{10}{3}\rightarrow(3)$ $\frac{eq(2)}{eq(1)}\Rightarrow\frac{n_{c_2}x^{n-2}y^2}{n_{c_1}x^{n-1}y}=\frac{30}{135}$ $\frac{n-1}{2}\left(\frac{y}{x}\right)=\frac{2}{9}\rightarrow(4)$ $\frac{eq(3)}{eq(2)}\Rightarrow\frac{n-2}{3}\left(\frac{y}{x}\right)=\frac{10}{3\times30}$ $\left(\frac{n-2}{3}\right)\left(\frac{y}{x}\right)=\frac{1}{9}\rightarrow(5)$ $\frac{(4)}{(5)}\Rightarrow\frac{\frac{n-1}{2}\left(\frac{y}{x}\right)}{\left(\frac{n-2}{3}\right)\left(\frac{y}{x}\right)}=\frac{2}{9}\times\frac{9}{1}$

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$$\frac{(n-1)^2}{2(n-2)} = 2$$

$$3n-3 = 4n-8$$

$$5 = n \quad in(1),(2),(3)$$

$$5_{c_1} x^4 y = 135, \qquad 5_{c_2} x^3 y^2 = 30, \qquad 5_{c_3} x^2 y^3 = \frac{10}{3}$$

$$x^4 y = 27 \qquad x^3 y^2 = 3 \qquad x^2 y^3 = \frac{1}{3}$$

Put $y = \frac{x}{9}$

$$x^5 = 27 \times 9$$

$$x = 3$$

$$y^2 = \frac{1}{9}$$

$$y = \frac{1}{3}$$

$$6\left(h^3 + x^2 + y^2\right) = 6\left(5^3 + 3^2 + \frac{1}{3}\right)$$

$$= 6\left(125 + 9 + \frac{1}{3}\right)$$

$$= \cancel{6}^2\left(\frac{375+27+1}{\cancel{3}}\right) \quad \Rightarrow 2(403) = 806$$

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